Figures Bundle

Decoupling Cyclone-Induced Saline Inundation from Agronomic Flooding in Sentinel-1/2 Rice Phenology Retrieval: A Multi-Year Bay-of-Bengal Coastal Framework (2017-2024)

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All figures are embedded at publication size (6.0-inch column width) from PNGs in figures/ regenerated from analysis/results/real_v21/. Each figure carries a short caption immediately below the image.

Figure 1

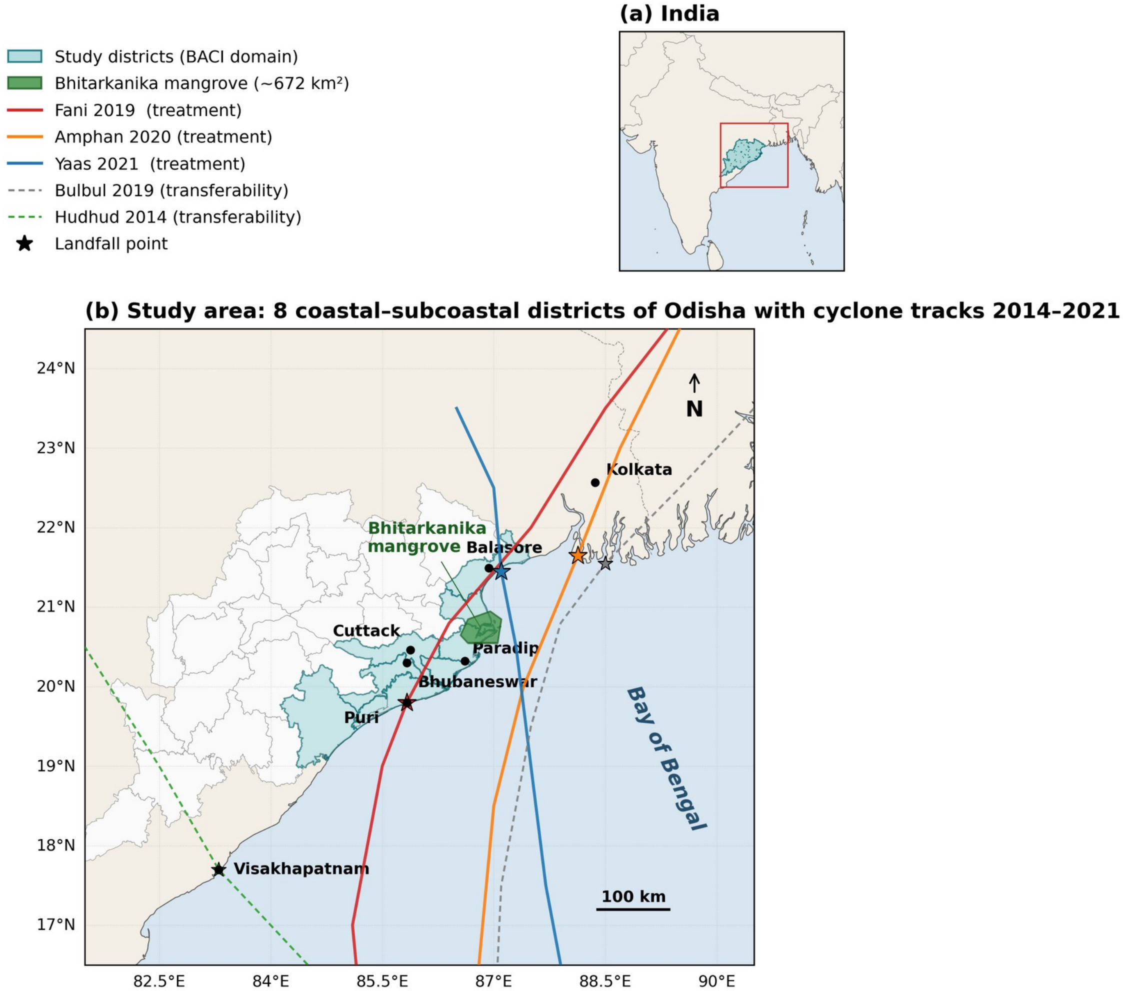


Figure 1. Study area: coastal and inland Odisha BACI districts. Treated coastal districts (Balasore, Bhadrak, Kendrapara, Jagatsinghpur, Puri) and inland control districts (Dhenkanal, Angul, Cuttack), with IBTrACS cyclone tracks for Fani (May 2019), Bulbul (Nov 2019), Amphan (May 2020), and Yaas (May 2021).

Figure 1B

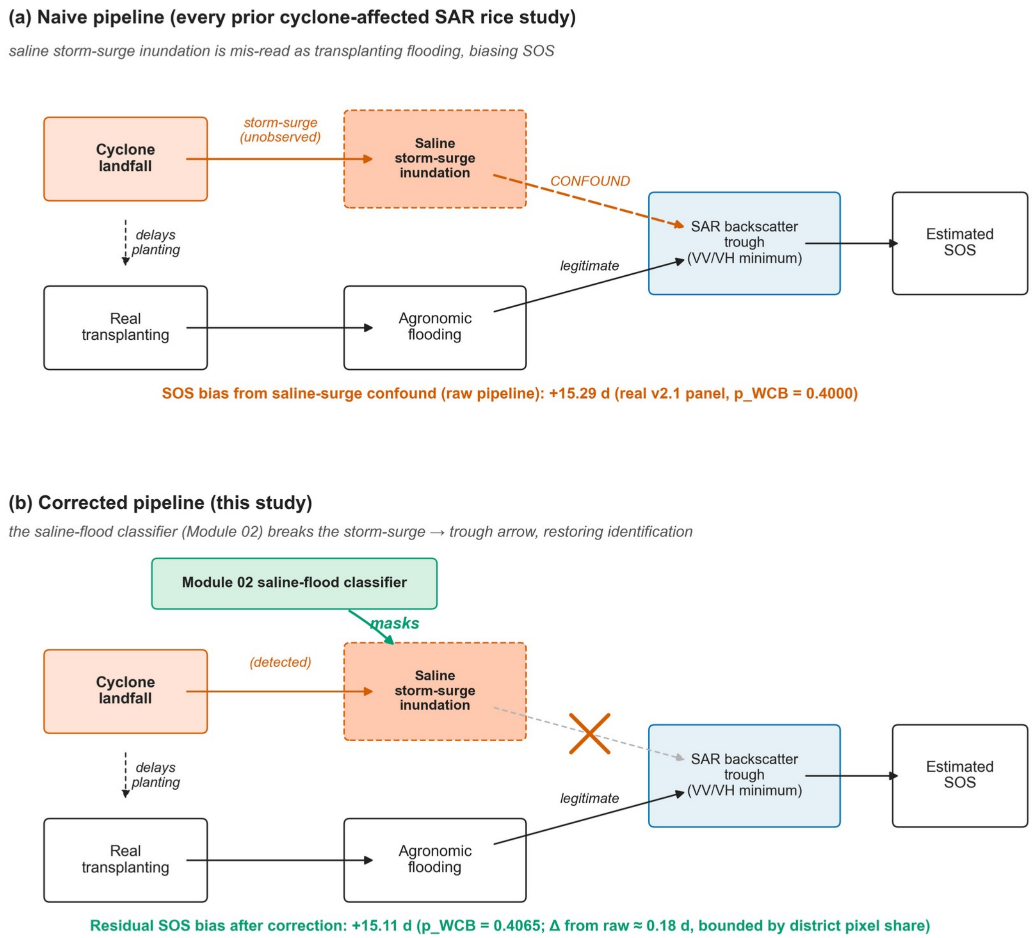


Figure 1B. Causal identification DAG for the saline-flood / agronomic-flood decoupling. The cyclone landfall is a single exogenous shock that opens parallel legitimate (transplanting flooding) and confounding (saline storm-surge) pathways into the same SAR backscatter trough; the Module 02 saline-flood classifier intercepts the confounding pathway.

Figure 2

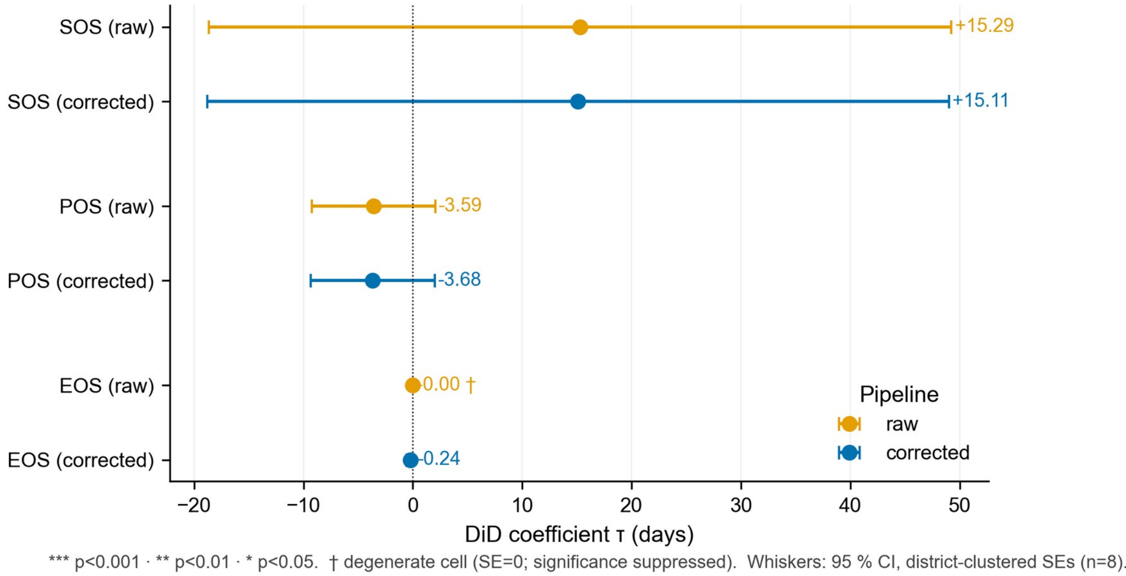
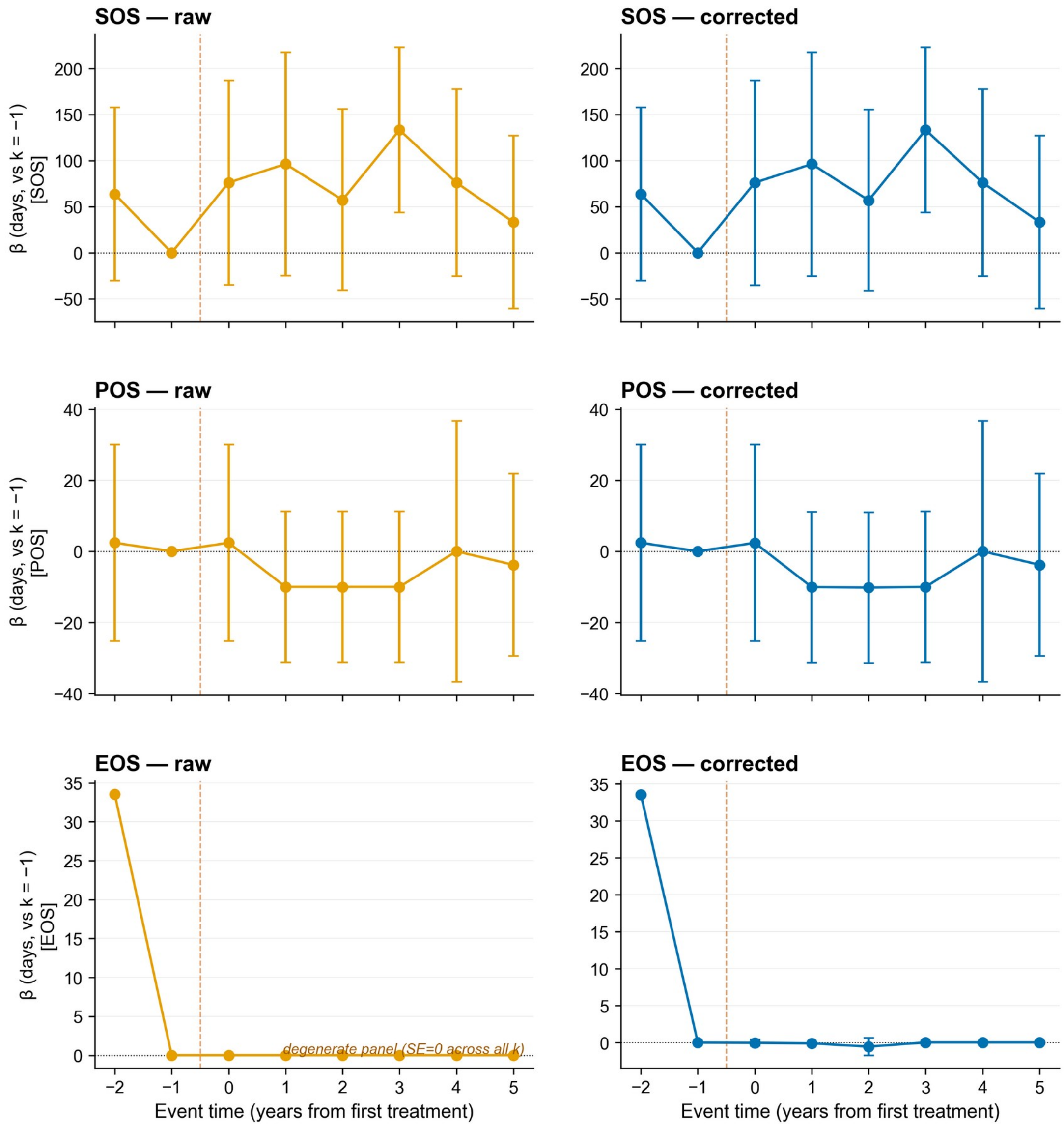


Figure 2. TWFE-DiD coefficients (raw vs. classifier-corrected panel). Point estimates with district-clustered (CR1) standard-error bars and 95% wild-cluster restricted bootstrap intervals for SOS, POS, and EOS, raw and corrected pipelines on the real v2.1 panel.

Figure 3



Reference period k = −1 (year before first treatment landfall, 2018). Whiskers: 95 % CI, district-clustered SEs.

Figure 3. Event-study coefficients on the real v2.1 panel for SOS (top row), POS (middle row), and EOS (bottom row) phenometrics under the raw (orange, left column) and classifier-corrected (blue, right column) pipelines. Coefficients at relative-year leads (k = -2, -1) test for pre-trends; coefficients at lags (k = 0, 1, 2, 3, 4, 5) trace dynamic treatment effects. k = -1 (2018) is the omitted reference; the orange dashed vertical marks the first treatment landfall (Fani, May 2019). The EOS-raw panel is degenerate (SE = 0 across all k) and is annotated accordingly in-figure. Whiskers: 95 % CI computed from district-clustered (CR1) standard errors.

Figure 4

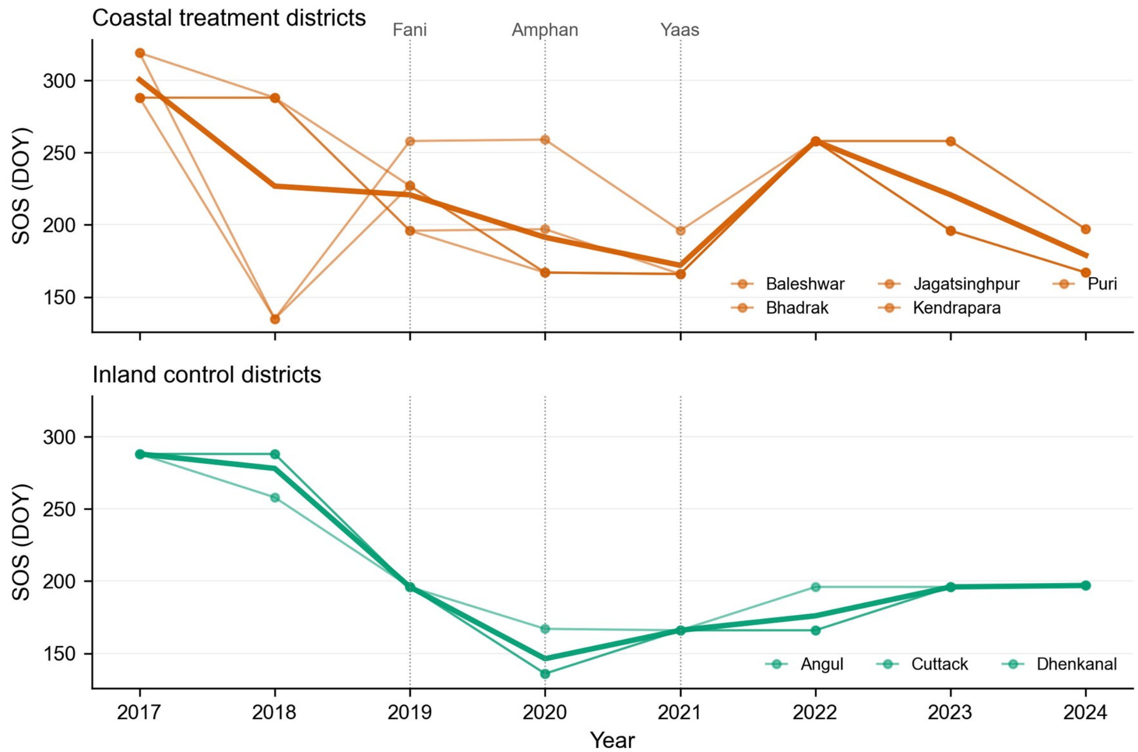


Figure 4. Per-district SOS trajectories on the classifier-corrected pipeline, Kharif 2017-2024. Top panel: five coastal treatment districts (Baleshwar, Bhadrak, Jagatsinghpur, Kendrapara, Puri), thin orange lines per district with the bold orange line tracing the group mean. Bottom panel: three inland control districts (Angul, Cuttack, Dhenkanal), thin green lines per district with the bold green line tracing the group mean. Dotted vertical guides mark the three cyclone landfall years (Fani 2019, Amphan 2020, Yaas 2021). Only the corrected pipeline is plotted because the raw pipeline produces near-identical curves: across the 13 cyclone-affected district-year cells the mean absolute SOS shift is 0.245 ± 0.272 days (range 0.04-1.01 d), with the largest single correction in Bhadrak 2021 (Cyclone Yaas, -1.01 d). Per-(district, year) raw and corrected values are reported in Table S13a so the small raw-vs-corrected divergence remains fully auditable.

Figure 5

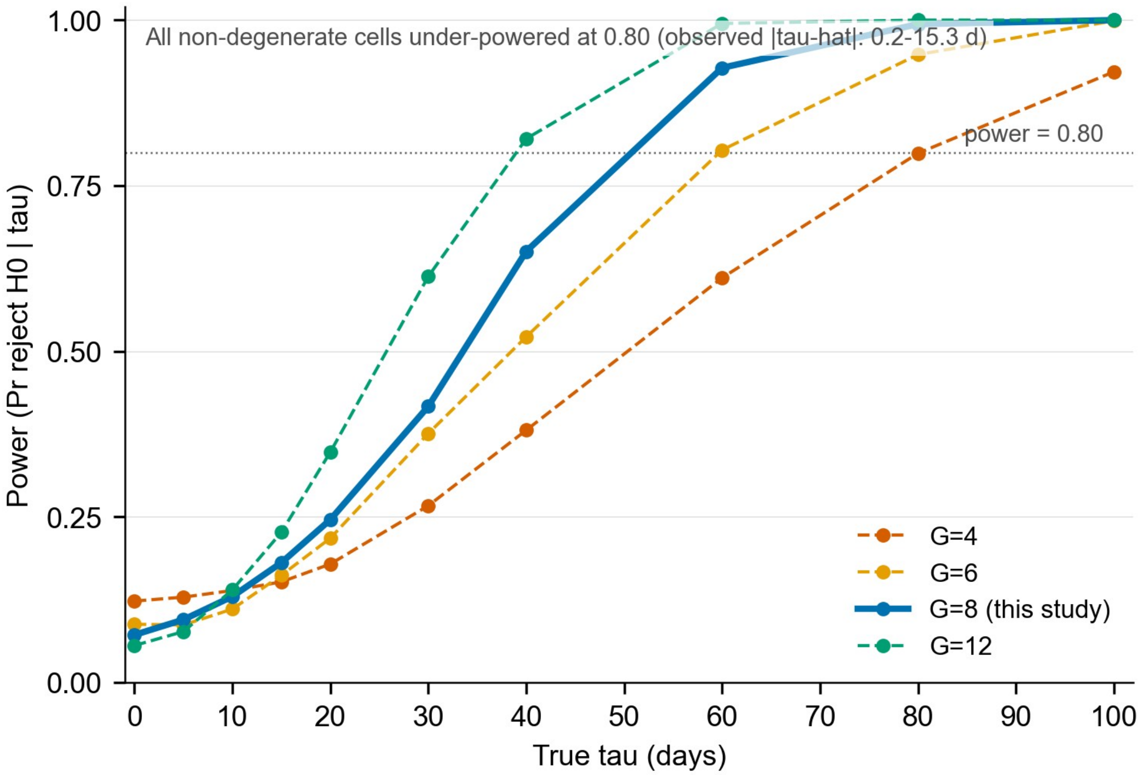


Figure 5. Empirical power curves from the Monte-Carlo simulation (Note S4). Rejection rate of H0 against true effect size tau over 999 replications per grid point at G = {4, 6, 8, 12} with empirical variance components sigma_u = 13.06 d, sigma_t = 41.75 d, sigma_epsilon = 31.28 d estimated from the real v2.1 corrected-SOS panel. At G = 8 (this study) power >= 0.80 is reached only for tau >= 60 d; the type-I rate under H0 is 0.07, close to nominal 0.05.

Figure 6

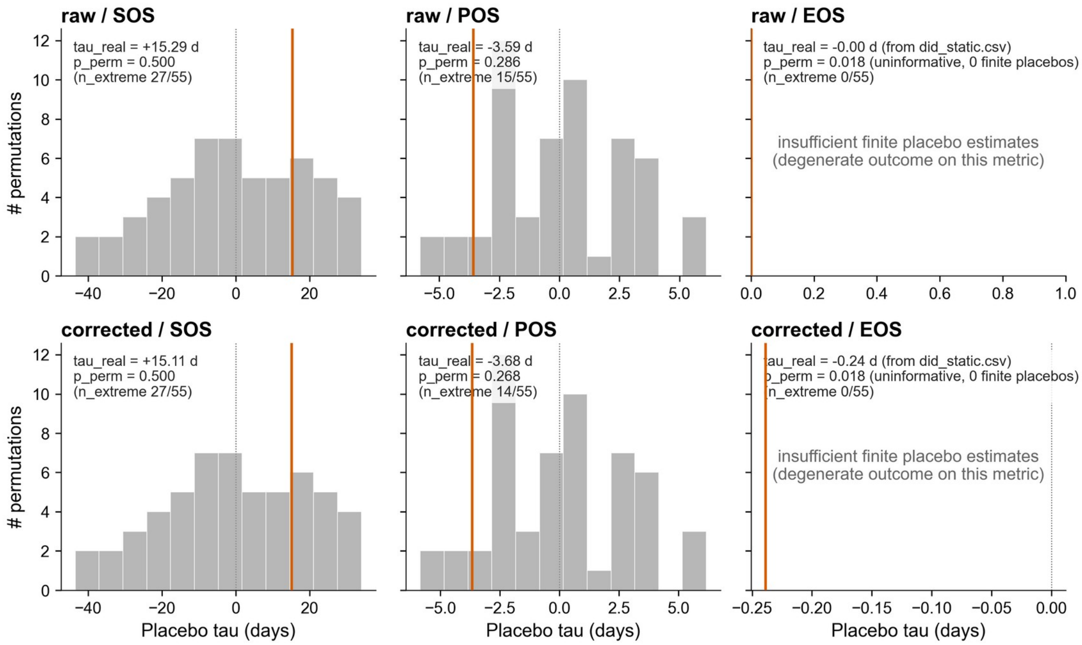


Figure 6. In-space placebo distributions (donor-swap, 56 permutations). Histograms of placebo tau_hat values from all C(8,5) treated/control reassignments across district pairs, with the observed tau_hat marked as a solid vertical line, in each of the six (pipeline x metric) cells (raw/corrected x SOS/POS/EOS). Both raw/EOS and corrected/EOS cells return p_perm = 0.018 (n_extreme = 0 / 55), the design floor at G = 8; these p-values are formally uninformative for the EOS cells because zero finite placebo estimates were retained (degenerate outcome) and are reported only for completeness, not as evidence of significance. The SOS and POS cells return p_perm = 0.286-0.500, consistent with the small-G null result.

Figure S1

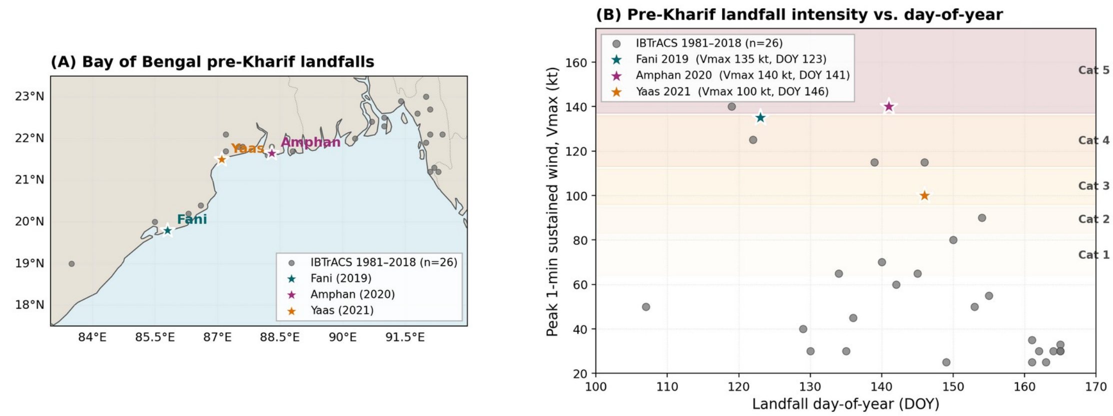


Figure S1. Cyclone climatology for the Bay of Bengal, 1980-2024. Annual landfall counts in the 50-km IBTrACS buffer around the coastal Odisha districts, stratified by Saffir-Simpson category and pre-monsoon vs. post-monsoon timing.

Figure S2

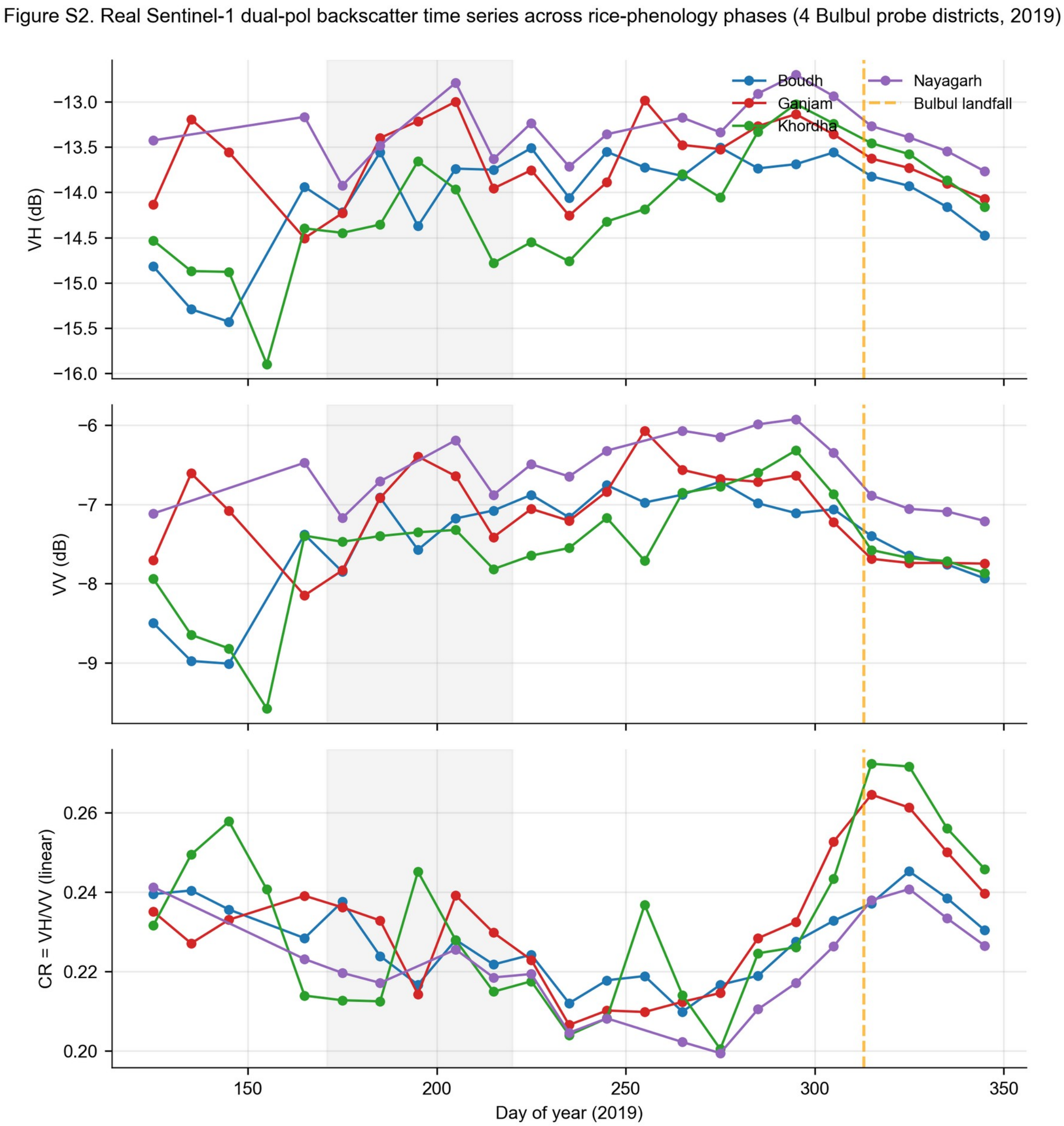


Figure S2. Sentinel-1 RTC dual-polarisation VH/VV/CR backscatter signatures across four phenological phases (pre-canopy, early canopy, peak canopy, event window) for the four Bulbul out-of-sample probe districts (Boudh, Ganjam, Khordha, Nayagarh), retrieved from Microsoft Planetary Computer.